

Assignment SU6: Browser Navigation Using Recursion and a Stack

Objective:

The objective of this assignment is to develop and test a Java program that simulates browser page navigation using recursion and a stack. The program will maintain a history of visited pages and use a stack to trace recursive navigation and backtracking. This will help you understand how stacks can be used to model the last-in, first-out (LIFO) behaviour of browser history while reinforcing recursive problem-solving.

Instructions:

1. Implement the BrowserHistory class

2. The class should maintain a stack containing the pages visited by the user.

3. Implement the recursive method visitPages()

This method should recursively process a sequence of web pages. Each page visited must be pushed onto the stack. The method should continue until all pages in the sequence have been processed.

4. Implement the recursive method goBack() This method should simulate the browser's Back operation recursively. At each call, the current page should be removed from the stack and the previous page should become the current page. The recursion must stop when the history stack contains only the first page or becomes empty.

5. Implement a method displayHistory()

Display the pages currently stored in the stack in an appropriate order. Clearly indicate the current page.

6. Develop the Test Program (BrowserTest.java)

- a. Prompt the user to enter the number of web pages to visit.
- b. Ask the user to enter each page name or URL.
- c. Store the entered pages in an array or ArrayList.
- d. Call visitPages() to recursively add the pages to the browser history stack.
- e. Display the complete browser history after all pages have been visited.
- f. Ask the user how many times they want to press the Back button.
- g. Call goBack() recursively to remove the requested number of pages, where possible.
- h. Display the browser history after navigating backwards.
- i. Display a trace showing each page pushed onto and popped from the stack.

7. Testing and Validation

- a. Test your program with different numbers of pages, for example: "Google", "YouTube", "Wikipedia".
- b. Test the program with only one page.
- c. Test the Back operation when the user requests more Back operations than the number of pages available.

- d. Confirm that every recursive visit pushes exactly one page onto the stack.
- e. Confirm that every recursive Back operation pops the correct page from the stack.
- f. Ensure that the stack correctly follows the Last-In, First-Out (LIFO) principle.
- g. Verify that the program handles an empty history appropriately.
- h. Create a Command Prompt (CMD) session to compile and run your complete Java files.

8. Required Java Files

Submit the following two Java source files:

- BrowserHistory.java – contains the BrowserHistory class, stack operations, and recursive methods.
- BrowserTest.java – contains the main() method and is responsible for user interaction and testing.

9. Expected Learning Outcomes

- Demonstrate an understanding of the Stack data structure and LIFO behaviour.
- Implement recursion to solve a practical navigation problem.
- Trace recursive calls using stack operations.
- Identify and correctly implement base cases for recursive methods.
- Test and validate a Java program using multiple input scenarios.

Important:

Your program must use recursion for both visiting pages and navigating backwards. Do not replace the required recursive methods with loops. The stack must be used to demonstrate the LIFO behaviour of browser history.